

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Buvanesh Y
Register Number	192372367

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

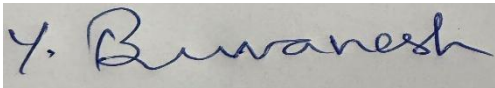
Question Count: 5

Code of Conduct

I Buavnes Y (192372367) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Type-1 vs Type-2 Hypervisors for Mission-Critical Cloud Datacenters:

Type-1 (bare-metal) runs directly on hardware — VMware ESXi, Microsoft Hyper-V, KVM.

Type-2 (hosted) runs atop a host OS — VMware Workstation, VirtualBox.

Feature	Type-1 Hypervisor (Bare Metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on hardware	Runs on top of an operating system
Performance	Very high, minimal overhead	Lower due to host OS overhead
Security	More secure, smaller attack surface	Less secure — host OS is an additional vulnerability
Resource Management	Direct access to CPU, memory, storage	Depends on host OS for scheduling and resource allocation
Reliability	Highly reliable for enterprise workloads	Suitable mainly for testing and development
Live Migration	Supported (vMotion, Hyper-V Live Migration)	Not supported
High Availability	Built-in HA and fault tolerance clustering	Absent
Examples	VMware ESXi, Microsoft Hyper-V, Xen, KVM	Oracle VirtualBox, VMware Workstation, Parallels

Analysis:

Performance: Type-1 hypervisors deliver near-native hardware performance because VMs communicate directly with physical resources — CPU, memory, and I/O — without routing through a host OS. Type-2 adds an extra scheduling layer: every guest instruction must pass through both the hypervisor and the host OS, introducing measurable latency that compounds at high VM density.

Security: Type-1 has a deliberately minimal trusted computing base — purpose-built software with no general-purpose OS beneath it. Type-2 inherits the full vulnerability surface of the host OS; a single kernel exploit can compromise every VM running above it. Regulated environments (PCI-DSS, HIPAA) mandate minimizing the TCB, which only Type-1 satisfies.

Manageability: Enterprise Type-1 platforms (VMware vSphere, Hyper-V with SCVMM) provide live migration, distributed resource scheduling, high availability clustering, storage vMotion, and role-based access control — all managed centrally across entire host clusters. Type-2 hypervisors are single-node tools with no native clustering or centralized management.

Justification

For mission-critical cloud applications, **Type-1 hypervisors are the definitive choice** because they provide:

- **Higher performance** — direct hardware access eliminates host OS overhead, critical for latency-sensitive workloads like financial transactions or real-time analytics
- **Stronger security** — minimal attack surface with no host OS layer for adversaries to exploit
- **Better fault tolerance** — built-in HA automatically restarts VMs on surviving hosts within seconds of a hardware failure
- **Enterprise scalability** — live migration, DRS, and cluster-wide management support hundreds of VMs per host with no downtime
- **Compliance readiness** — minimal TCB satisfies audit requirements under PCI-DSS, HIPAA, and SOC 2

Type-2 hypervisors remain useful for developer workstations and isolated test labs, but have no place in production cloud infrastructure.

2. Workload Isolation Across Finance, Healthcare, and Student Portals

How virtualization architecture supports isolation:

Each tenant workload runs in a separate VM, enforcing hardware-level memory boundaries — the hypervisor's memory management unit (MMU) virtualization ensures one VM cannot read another's address space. Separate virtual NICs, VLANs, and virtual switches segment network traffic so finance packets never traverse the same broadcast domain as student traffic. Storage isolation is achieved through dedicated virtual disk images and, in stricter deployments, separate LUNs with zoning.

For compliance-heavy tenants (finance under PCI-DSS, healthcare under HIPAA), dedicated resource pools and policy-based network micro-segmentation (e.g., VMware NSX) enforce that even lateral East-West traffic between co-resident VMs is inspected and restricted.

Two major attack surfaces:

1. VM Escape. A vulnerability in the hypervisor itself (e.g., in the virtual device emulation layer) can allow a malicious guest to break out of its VM and gain access to the host or sibling VMs. This is rare but catastrophic — Cloudburst (VMware) and VENOM (QEMU's floppy controller) are historical examples. Mitigations include disabling unused virtual device emulators and keeping hypervisor patches current.

2. Shared Memory Side-Channels. Co-located VMs share physical CPU caches and memory buses. Spectre and Meltdown demonstrated that timing attacks on shared L3 cache or branch predictor state can leak secrets across VM boundaries without any hypervisor vulnerability. Mitigations (retpoline, KPTI, core scheduling) reduce but don't eliminate this risk, and they carry a performance cost.

Protection Measures:

- Keep hypervisors updated.
- Enable Secure Boot and TPM.
- Use network segmentation.
- Apply least-privilege access control.

3. Analyze VM Host Utilization Snapshot

Given Data

Metric	Value	Status
CPU Utilization	92%	Critical
Memory Utilization	88%	High
Storage I/O Latency	35 ms	Degraded
Average VM Boot Time	170 seconds	Severe

Likely Bottlenecks

CPU Bottleneck

- CPU usage at 92% leaves almost no headroom for burst demand.
- VMs experience high CPU ready time — waiting for a physical core to become available.
- Applications become sluggish; real-time and transactional workloads are directly impacted.
- Over-provisioned vCPUs across VMs worsen contention even further.

Memory Bottleneck

- Memory at 88% pushes the hypervisor toward ballooning and swapping.
- Ballooning reclaims guest memory by pressure; swapping writes VM memory pages to disk — both degrade performance significantly.

- Swapped memory pages hitting an already-congested storage subsystem create a compounding bottleneck.
- Applications may crash or stall if the hypervisor cannot reclaim memory fast enough.

Storage Bottleneck

- 35 ms I/O latency far exceeds acceptable thresholds (under 10 ms for SSD, under 20 ms for HDD-backed storage).
- Indicates disk queue saturation — too many concurrent read/write requests from competing VMs.
- Every disk-dependent operation is slowed: application data access, page file reads, VM snapshot I/O.
- A noisy-neighbor VM monopolizing I/O bandwidth is a common cause in high-density deployments.

Slow Boot Time

170 seconds indicates a combination of all three bottlenecks simultaneously:

- Storage congestion delays kernel and application file loading during boot.
- CPU contention means the booting VM cannot get scheduled promptly.
- Memory pressure forces the hypervisor to swap pages that the new VM immediately needs.
- Too many VMs attempting concurrent operations amplifies all three effects at once.

Corrective Actions

Area	Action
CPU	Add physical CPU cores or migrate VMs to underutilized hosts via live migration
CPU	Audit and reduce over-allocated vCPUs; target steady-state utilization below 70%
Memory	Increase physical RAM on the host
Memory	Reclaim idle VM memory using ballooning metrics; reduce over-allocated VM memory
Storage	Upgrade datastore to NVMe or SSD-backed storage to bring latency below 5 ms
Storage	Enable Storage I/O Control (SIOC) to prevent noisy-neighbor VMs from saturating I/O
Storage	Separate boot/swap datastores from application data datastores
General	Reduce VM density — distribute workloads across additional hosts
General	Enable Distributed Resource Scheduler (DRS) for automated load balancing

Area	Action
General	Implement continuous monitoring with threshold alerts at 75% CPU, 80% memory

Conclusion

The host is critically overloaded across all three resource dimensions — CPU, memory, and storage — and they are reinforcing each other. Memory pressure drives swapping to an already-saturated disk subsystem, which compounds storage latency, which in turn delays every VM operation including boot. CPU contention prevents any individual VM from recovering quickly. The 170-second boot time is the clearest symptom of this cascade. Immediate corrective action requires storage hardware upgrades, VM migration to redistribute load, and physical RAM expansion — treating any single dimension in isolation will not resolve the overall degradation.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter

Feature	Traditional Datacenter	Software Defined Datacenter
Compute	Physical servers	Virtualized compute resources
Storage	Dedicated storage devices	Software-defined storage
Networking	Hardware switches	Software-defined networking (SDN)
Provisioning	Manual	Automated
Scalability	Limited	Dynamic
Resource Allocation	Static	Elastic

Compute Virtualization

- Creates virtual machines.
- Enables rapid deployment.
- Supports live migration and auto scaling.

Storage Virtualization

- Pools storage into a centralized resource.

- Improves availability.
- Simplifies backup and disaster recovery.

Network Virtualization

- Creates virtual networks independent of hardware.
- Enables micro-segmentation.
- Provides software-based traffic management.

How SDDC Improves Agility

- Faster deployment of services.
- Automated provisioning.
- Better resource utilization.
- Simplified management.
- Reduced operational costs.
- Easier disaster recovery.

Conclusion

SDDC offers greater flexibility, automation, scalability, and efficiency compared to traditional datacenters.

5. Multi-Cloud Federation Identity Problem

Problem Analysis:

A multi-cloud federation fails because each cloud provider uses different identity systems:

- Different user IDs
- Different authentication methods
- Different authorization policies
- Duplicate user accounts

This results in:

- Authentication failures
- Poor user experience

- Increased administrative effort
- Security risks

Secure Identity Federation Design:

Implement a centralized **Identity Provider (IdP)**.

Components:

- Identity Provider (IdP)
- Service Providers (Cloud A, Cloud B, Cloud C)
- Federation protocols:
 - SAML
 - OAuth 2.0
 - OpenID Connect (OIDC)

Privacy Protection

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Encrypted authentication tokens
- Minimal attribute sharing
- End-to-end encryption
- Regular auditing and logging

Benefits

- One identity across all cloud providers.
- Secure authentication.
- Improved privacy.
- Reduced password management.
- Easier administration.

- Better compliance with security standards.

Conclusion:

Using identity federation with **SAML/OAuth 2.0/OpenID Connect**, along with **SSO, MFA, RBAC, encryption, and privacy-preserving attribute sharing**, provides secure and seamless access across multiple cloud providers while protecting user identities and sensitive data.